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Course: Calculus Beta

Assignment: MyLab Math - Uge 13

Find all the local maxima, local minima, and saddle points of the function.

$$f(x,y) = x^2 + xy + y^2 + 2x - 2y + 6$$

The local extrema and saddle points in the interior of the domain of a function occur at critical points of the function. An interior point of the domain is a critical point if both f_x and f_y are zero or if one or both derivatives do not exist.

Find the partial derivative with respect to x .

$$f_x = 2x + y + 2$$

Find the partial derivative with respect to y .

$$f_y = x + 2y - 2$$

Both derivatives exist everywhere in the domain of $f(x,y)$. Set each derivative equal to zero and solve the system of equations for x and y .

$$2x + y + 2 = 0$$

$$x + 2y - 2 = 0$$

Solve the system by substitution. Solve the equation $f_y = 0$ for x .

$$x + 2y - 2 = 0$$

$$x = -2y + 2$$

Substitute $-2y + 2$ for x in the equation for f_x , and solve for y .

$$2(-2y + 2) + y + 2 = 0$$

$$y = 2$$

Substitute 2 for y in the expression for x and simplify.

$$\begin{aligned} x &= -2(2) + 2 \\ &= -2 \end{aligned}$$

There is only one critical point at $(-2, 2)$ for $f(x,y) = x^2 + xy + y^2 + 2x - 2y + 6$.

The nature of the critical points can be tested using the second derivative test if the first- and second-order partial derivatives of f are continuous throughout a disk centered at point (a,b) and $f_x(a,b) = f_y(a,b) = 0$.

i. f has a local maximum at (a,b) if $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a,b) .

ii. f has a local minimum at (a,b) if $f_{xx} > 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a,b) .

iii. f has a saddle point at (a,b) if $f_{xx}f_{yy} - f_{xy}^2 < 0$ at (a,b) .

iv. The test is inconclusive at (a,b) if $f_{xx}f_{yy} - f_{xy}^2 = 0$ at (a,b) .

The expression $f_{xx}f_{yy} - f_{xy}^2$ is called the discriminant. Find each derivative in the expression. Differentiate f_x with respect to x to find f_{xx} .

$$f_x = 2x + y + 2$$

$$f_{xx} = 2$$

Differentiate f_y with respect to y to find f_{yy} .

$$f_y = x + 2y - 2$$

$$f_{yy} = 2$$

Differentiate f_x with respect to y to find f_{xy} .

$$f_x = 2x + y + 2$$

$$f_{xy} = 1$$

Evaluate the discriminant at $(-2, 2)$ to determine the nature of the critical point at that point.

$$\begin{aligned}f_{xx}f_{yy} - f_{xy}^2 &= (2)(2) - (1)^2 \\ &= 3\end{aligned}$$

At the point $(-2, 2)$, the discriminant is 3 and f_{xx} is 2. Since the discriminant is positive and f_{xx} is positive, the point is a local minimum.

The natures of all the critical points for $f(x, y) = x^2 + xy + y^2 + 2x - 2y + 6$ have been determined.

There are no local maxima for $f(x, y) = x^2 + xy + y^2 + 2x - 2y + 6$.

There is a local minimum at $(-2, 2)$. Find the value of the function at that point by substituting the values of the coordinates into the function.

$$\begin{aligned}f(x, y) &= (-2)^2 + (-2)(2) + (2)^2 + 2(-2) - 2(2) + 6 \\ &= 2\end{aligned}$$

There are no saddle points for $f(x, y) = x^2 + xy + y^2 + 2x - 2y + 6$.